

Summary: Solar

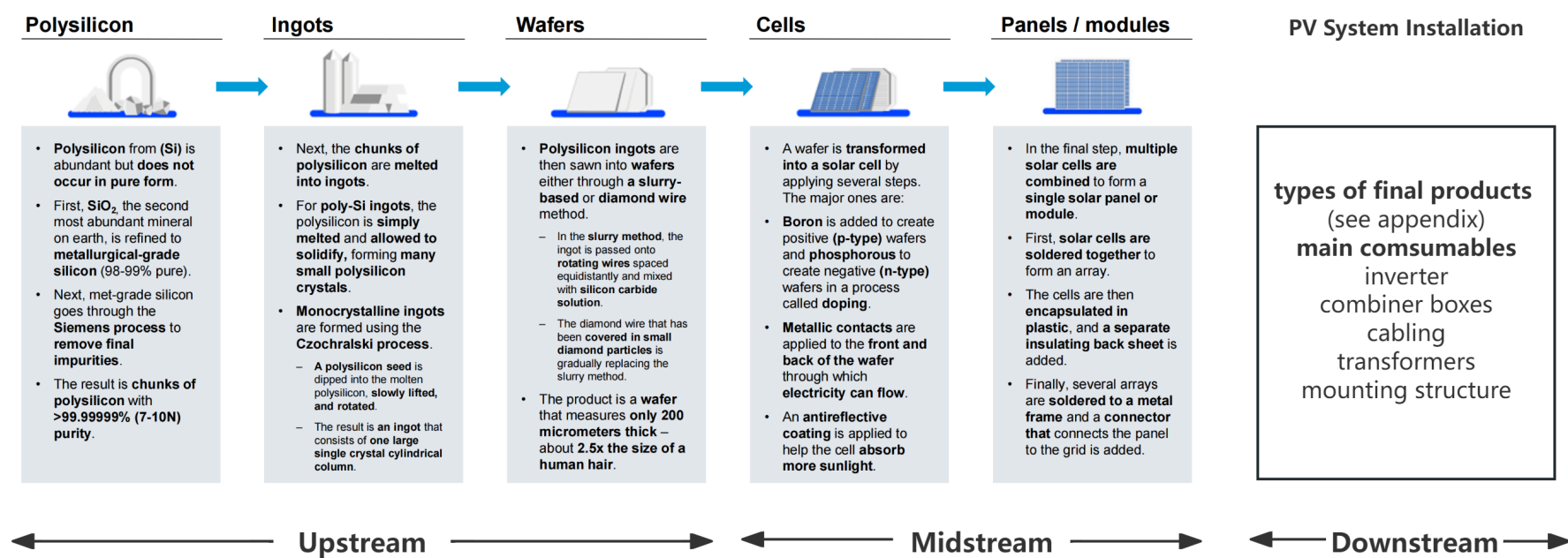


Key Takeaways:

- The upstream manufacturing of solar pv is dominated by Chinese firms, while the downstream installation tends to be more localized.
- PV cells are highly applicable to most regions, but the performance is compromised in dusty/hot environments, which requires adaptation technologies.
- For the whole PV system, one mismatch is the need of off-grid solar system in poor regions. Off-grid solar requires energy-storage batteries, which shares many similarities with EV batteries.
- There's little evidence showing technology mismatch between weak-grid vs strong-grid. Renewable energy generation adds additional instability issues, which are present in both weak and strong grid countries. Enabling technologies are important for dealing with the issue.
- China has been the primary driver of global solar PV cost reductions, making solar energy a more viable option for developing countries to pursue energy transitions (e.g., Africa, SIDS)
- Three key factors behind China's cost reduction in solar PV are particularly interesting: Local-adapted manufacturing, Learning-by-doing and Industry cluster
- Some thoughts of implications for other developing countries, say appropriate product, appropriate manufacturing model, appropriate entrepreneurship, and geopolitical cooperation in Belt & Road Initiative countries
- Appendix: a mind map of solar product types

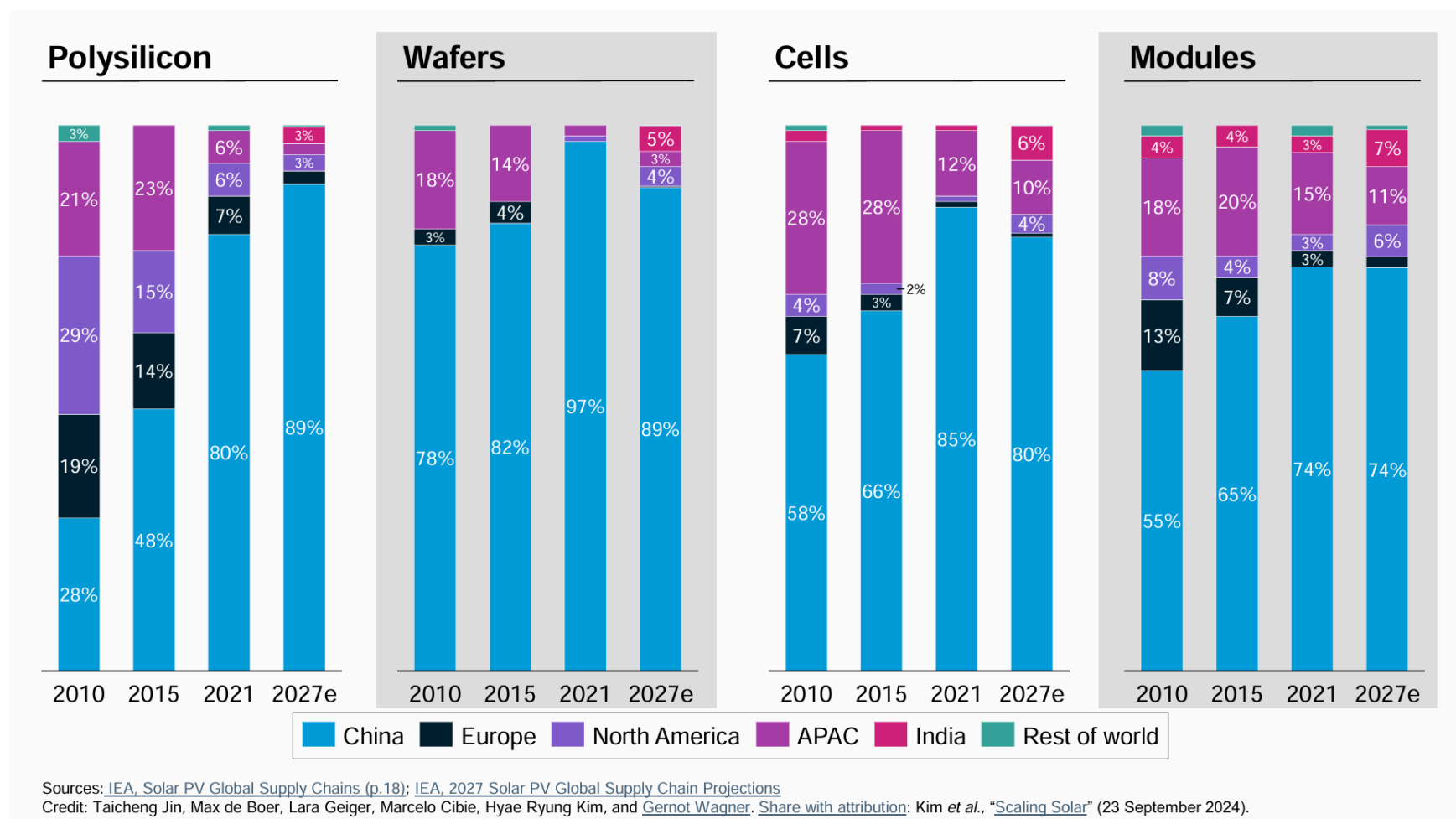
Notes: sentences with red color are new addition to the last report of solar (summary 0511)

1 Overview of Solar PV value chain



Source: PV Education, Refining Silicon; PV Manufacturing, Wafering; images from IEA, Solar PV Global Supply Chains.

China dominates over ~75% in almost every step of upstream & midstream PV value chain. And many leading Chinese manufacturers are vertically integrated across multiple steps (e.g., ingot/wafer/cell/module), but only a few are fully end-to-end (polysilicon → module).



While for downstream installation, it tends to be more localized. For example, Nigeria's solar market is import-dependent upstream but largely localized downstream. For modules, the country relies heavily on imports: by June 2025, Nigeria had imported 1,721 MW of panels from China, making it Africa's second-largest importer. By contrast, downstream activity is mostly handled by locally-registered firms: among the hundreds of listed installers in Nigeria (source: https://www.enfsolar.com/industry-directory?_gl=1*159svrw*_gcl_au*MTE1NTA2OTgzNS4xNzU5MDQ3MTc2), the vast majority are locally incorporated and headquartered or branched in Lagos, Abuja, or Kaduna. These companies typically cover project development, system design and installation, financing, and after-sales O&M. A minority of players operate via foreign registrations—for example, MBH Power lists a Singapore address for its group entity while maintaining a substantial Nigerian presence and offices in Lagos.

2 Technology Mismatch

2.1 PV cells

The most technology-intensive part of solar PV system is PV cells, which in general are highly modular, standardized, and technically applicable across most regions. However, there are also some examples of local adaptation requirements for PV cells.

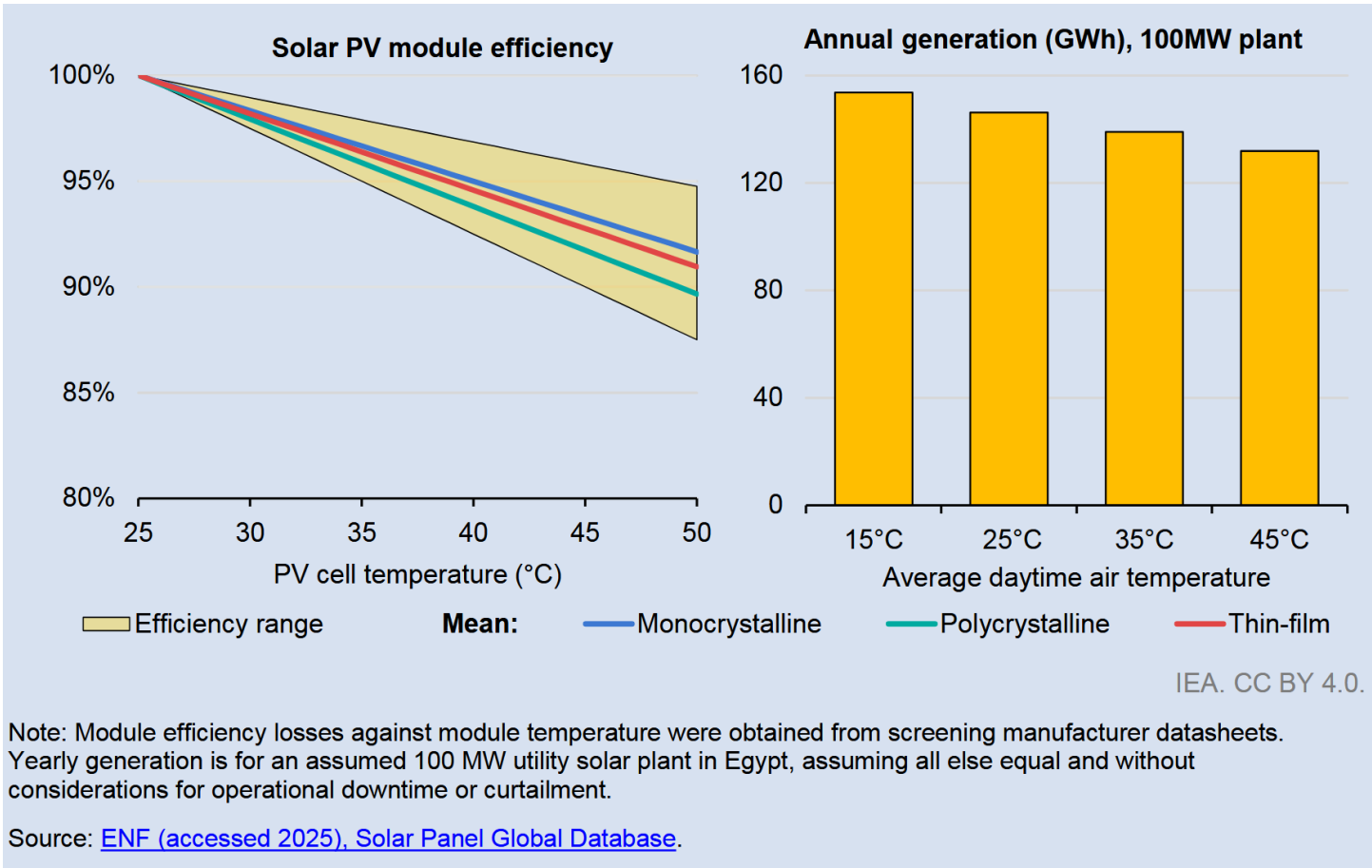
Ex.1 dust and soiling can cause up to 30% energy loss if not managed. Examples of solutions are spray-and-brush cleaning systems in Morocco and tilted, ventilated mounting structures in India.

Ex.2 high ambient temperatures reduce module performance

The physical processes underpinning solar photovoltaic (PV) output are intrinsically impacted by temperature, such that PV cell performances decrease when PV cell temperatures rise above 25 °C. Higher solar cell temperatures lead to voltage drops within the cells that outweigh increases in current, causing power output to decrease. This effect is summarized in a linear relationship approximated by the cell temperature coefficient, measured as the percentage loss of maximum power output per degree Celsius. **Different photovoltaic cell technologies have different sensitivities, varying from -0.2% to -0.5% per extra 1 °C**, with **monocrystalline cells performing better** under higher temperature conditions. For example, an air temperature difference of +10 °C can decrease cell efficiency by 3.3% and annual output by 7.2 GWh for a 100 MW monocrystalline plant, all else equal.

The cell temperature depends on ambient conditions (temperature, wind speed, surrounding albedo, irradiation), cell materials, as well as module and plant design. Many of the locations with the highest solar potential also correspond to some of the warmest locations on the planet. Hence, the choice of location, adaption measures to local conditions, and mitigation of rising temperatures cannot be understated. Some measures have already been implemented while others are

the subject of active research. There is a lot focus on **improving cell design and materials**, for example with **N-type cells being less sensitive to temperature**. Some cooling methods are already well understood like **wind convection cooling** or **water cooling**, having been tested in Morocco or the United Arab Emirates amongst others, while more novel ones are being explored like **phase-change materials (PCM)**, **hybrid systems in Malaysia** or **reflective coatings in Australia**. Many remain to be widely adopted.



2.2 Off-grid vs On-grid

In terms of the whole solar PV system, one important mismatch is the need of off-grid solar system in remote and less developed regions where many households get no access to grid

- Two types of solar system:
 - (i) Off-grid systems: solar systems that provide electricity to households and villages that are not connected to the utility electricity network. e.g., Solar home system(SHS), mini-grids.
 - (ii) On-grid systems: In contrast, grid-connected systems feed directly into the utility network and are more cost-effective at scale. In many developed countries, the share of deployment is disproportionally large.
- Deployment share by country:

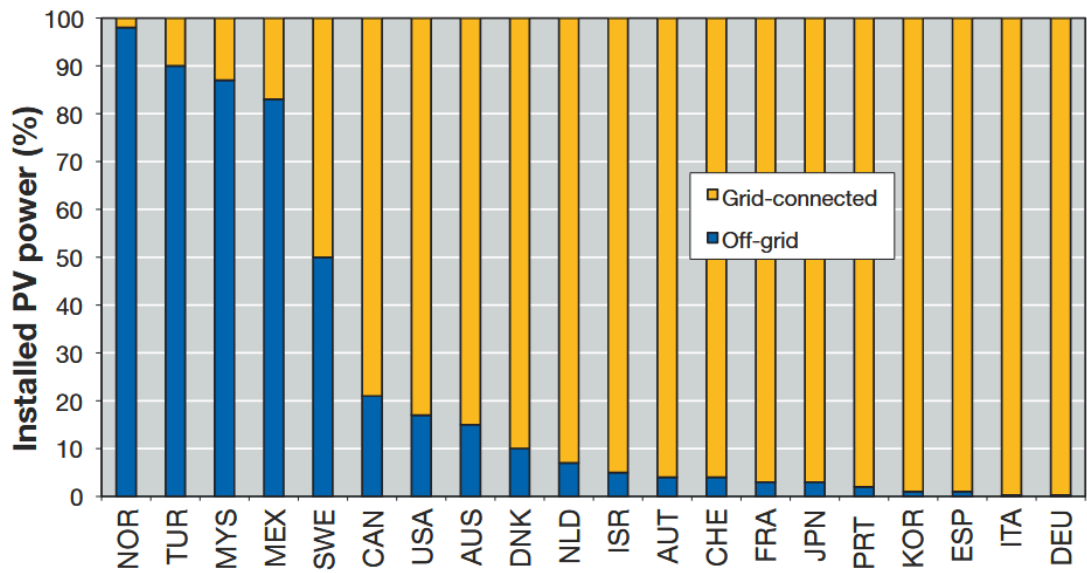
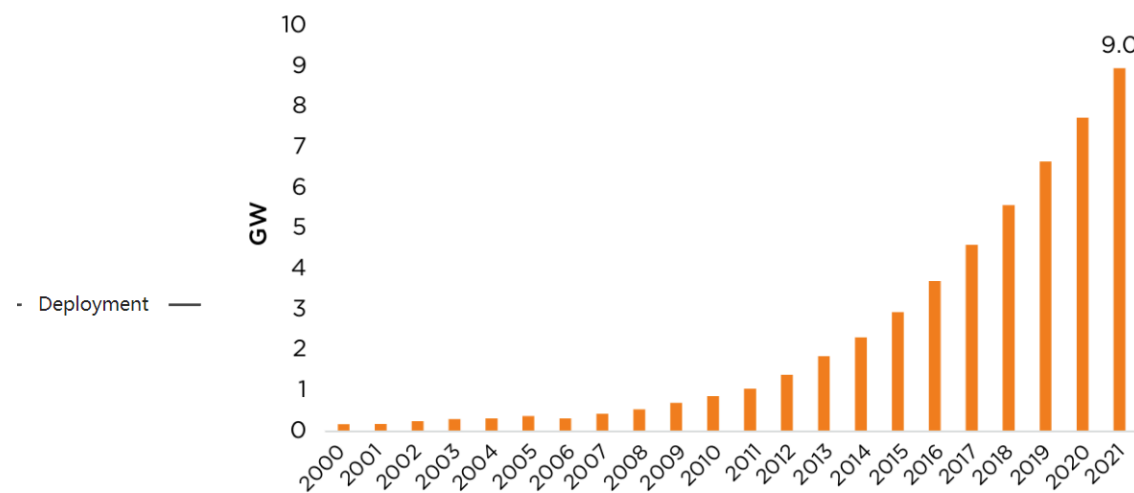


Figure 3a – Installed PV power in the reporting countries by application (%) in 2010

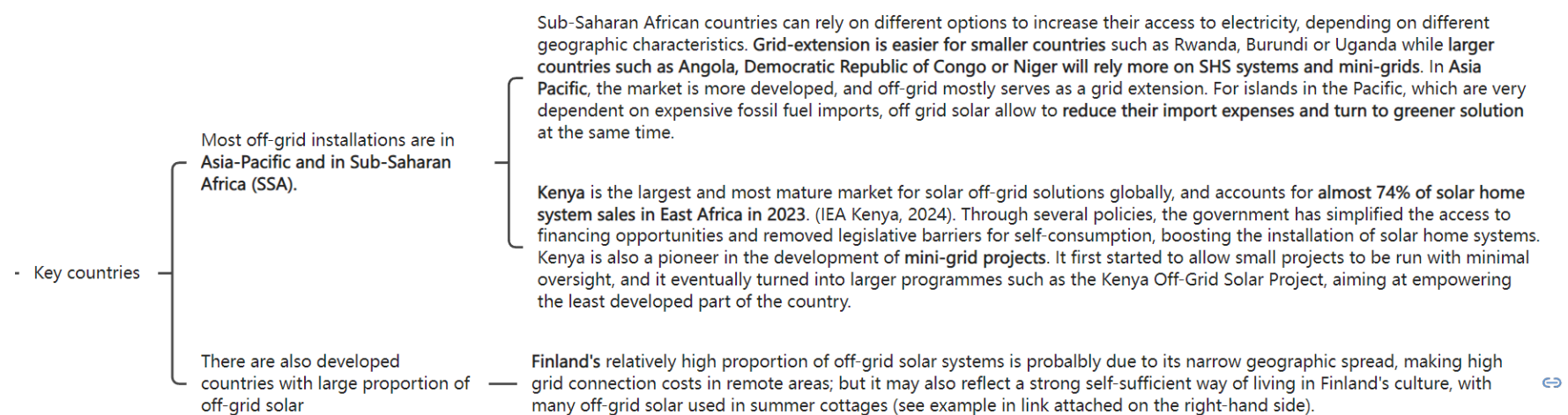
- Deployment scale and key user countries:

Fig 35. Cumulative off-grid solar PV capacity 2000-2021



Source: IEA, IRENA, IHS

Notes: Cumulative installation of off-grid solar is still limited, constituting less than 1% of total solar PV capacity. As still over 700 million people miss access to stable electric supply, off-grid PV remains a large market with nearly 500 million users worldwide in 2021.



Notes: Snapshot from the appendix "a mind map of different types of solar". Please see more details of solar product differentiation (features, deployment scales, and key user countries in the appendix.

- **One type of off-grid system is solar home system(SHS), which is increasingly popular in African countries, thanks to lower cost of pv modules and local-adapted business model.**
 - What is SHS? Solar Home Systems (SHS) combine the use of efficient lights (mostly LEDs) with charge controllers and batteries. With a small PV panel of only a few watts, essential services can be provided, such as lighting, phone charging and powering a radio or a small computer. Expandable versions of pico PV systems have entered the market and enabled starting with a small kit and adding extra loads later. They are mainly used for off-grid basic electrification, predominantly in developing countries.
 - What makes it suitable?
 - (i) Low-skill-requirement with **"plug-and-play" nature**, which allows users to set up and operate small solar kits without technical expertise. Typically, non-hardware "soft costs" -- like permitting, installation, and interconnection -- make up more than half the total cost of solar energy system. The invention of plug-and-play product allows fast installation and automatic permitting, interconnection and inspection. (news: https://www.energy.gov/eere/success-stories/articles/eere-success-story-plug-and-play-purchase-install-and-connect?nrg_redirect=249276) But this plug-and-play solar product was pioneered by CraftStrom Solar, a U.S. company, not developing countries.
 - (ii) Local-adapted business model

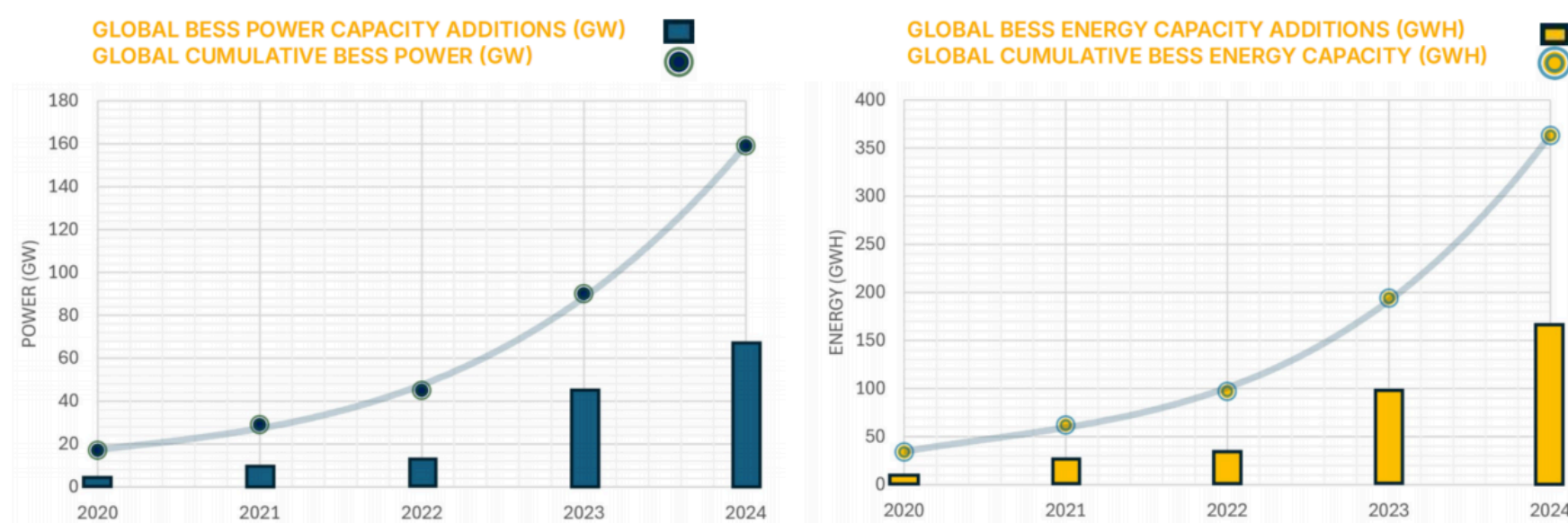
To make these systems affordable, many companies(e.g., M-KOPA) in Kenya use a **pay-as-you-go (PAYG)** model. Under this approach, users pay a small deposit and then make daily or weekly payments via mobile money platforms like M-Pesa(which is interesting to me, for it shows somewhat **"appropriate finance"** in Africa). Each payment unlocks continued access to the system. If a payment is missed, the system can be remotely disabled until the next payment is made. After completing all payments—usually over 12 to 24 months—the user owns the system outright.
 - (iii) Lower cost.

Africa's systems, typically under 100 W, are tiny compared to their counterparts in developed countries and require batteries and charge controllers to ensure stable output. As a result they have higher costs per kilowatt. But as prices of solar PV modules fall, SHS became an economic alternative for the 600 million Africans without a grid connection(IEA Solar PV in Africa, 2016)

- **Energy storage system of off-grid solar**

Off-grid solar today mostly uses lithium-ion batteries, especially lithium iron phosphate (LFP), with the supply chains, manufacturing steps, recycling pathways, and safety standards largely overlapping with those for EV batteries. The installation capacity of battery energy storage system (BESS) continues to exceed expectations, with 69 GW / 169 GWh of BESS installed in 2024. Li-ion batteries represented over **98%** of battery installations in 2024, partly because of the falling price of Li-ion batteries.

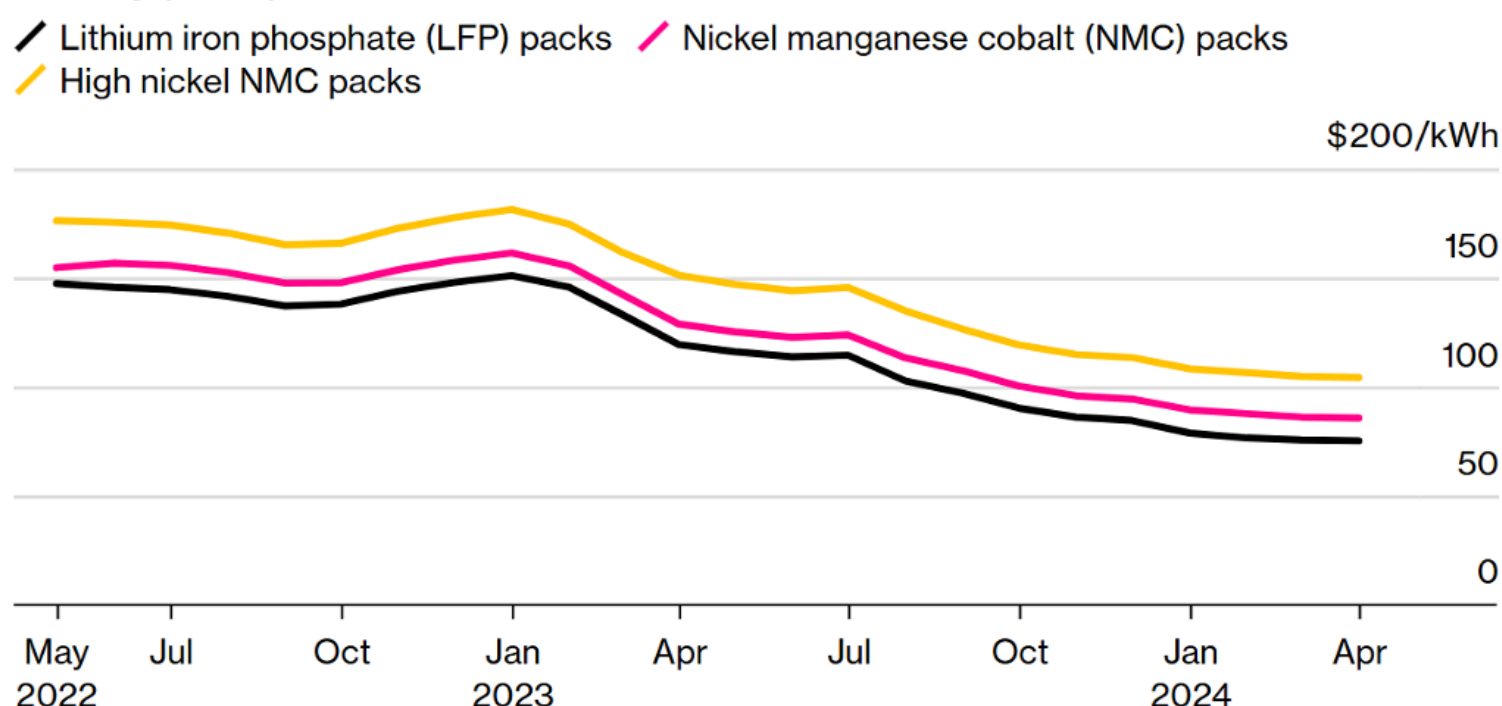
- Global BESS Deployment Soar



Source: [2024 Annual Battery Report - Download](#)

- Lithium-ion battery prices are dropping fast

Battery pack prices in China



Source: BloombergNEF, ICC Battery

Note: NMC = Nickel manganese cobalt and includes prices for NMC111, NMC532 and NMC 622 batteries. High-nickel NMC includes NMC811, NMC955 and NCA

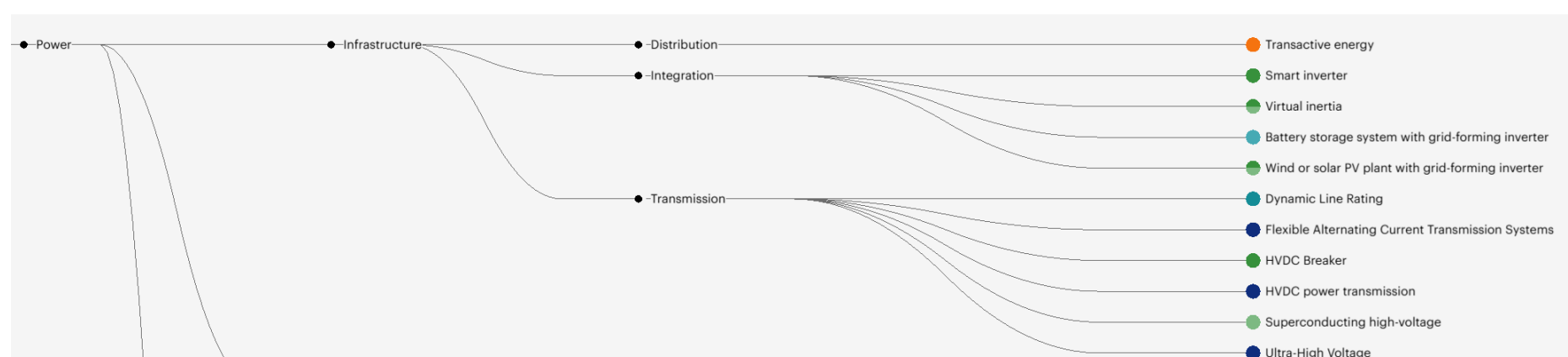
2.3 Weak-grid vs Strong-grid

Renewable energy generation needs converters to be grid-compliant, which leads to additional instability issues.

(Compared with conventional power system, converter-dominant system has less inertia, which means the system has less time to recover from sudden equipment failure). **It is especially challenging in weak grids**, typically featured with long transmission lines, high grid impedance, unfavorable grid structures (not meshed and interconnected, like Australia), and limited coverage. **The example of Nigeria demonstrates this concern.** Nigeria's national grid is very weak, with about 85 million Nigerians still lacking access to electricity, according to World Bank. Aging infrastructure is a primary concern, while underinvestment and vandalisms on transmission infrastructure exacerbates the issue. Frequent grid fluctuations and

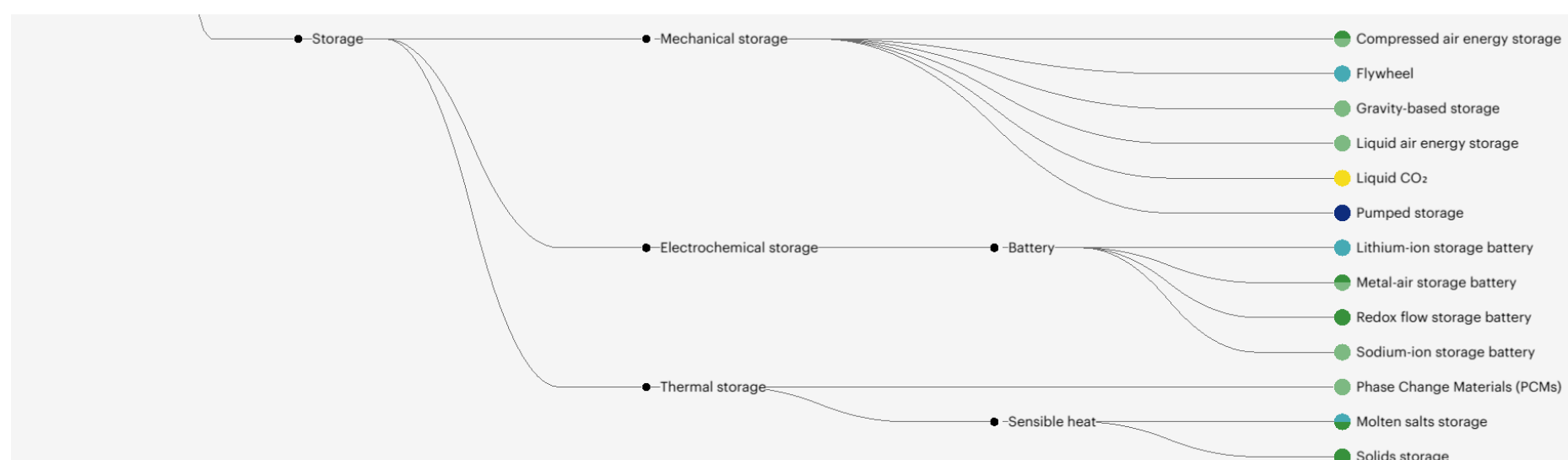
power surges can damage sensitive solar components, particularly inverters. According to the Transmission Company of Nigeria (TCN), the national grid collapsed 46 times between 2017 and 2020, exacerbating solar system failures. **However, the issue is not specific to developing countries, but also common for developed countries with high proportion of renewable energy generation.** For example, in Australia, with a high proportion of wind farms and rooftop solar, South Australia has become increasingly reliant on an interconnector across the Victorian border to stabilise its grid. When the interconnector failed in September, 2016, there was not enough synchronous generation turned on within SA's borders to arrest the rate of change of frequency, resulting in eventual blackout.

Enabling technologies are important to improve the performance of renewable energy. Infrastructure technologies such as virtual inertia (which emulates the inertial response of traditional synchronous machines using power electronics and fast control) and grid-forming converters (which are unlike conventional grid-following converters depending on existing grid signal, use instead advanced control and often energy buffers to actively set voltage and frequency to stabilize the grid), along with smart inverters, can help mitigate instability issues in converter-dominated grids.

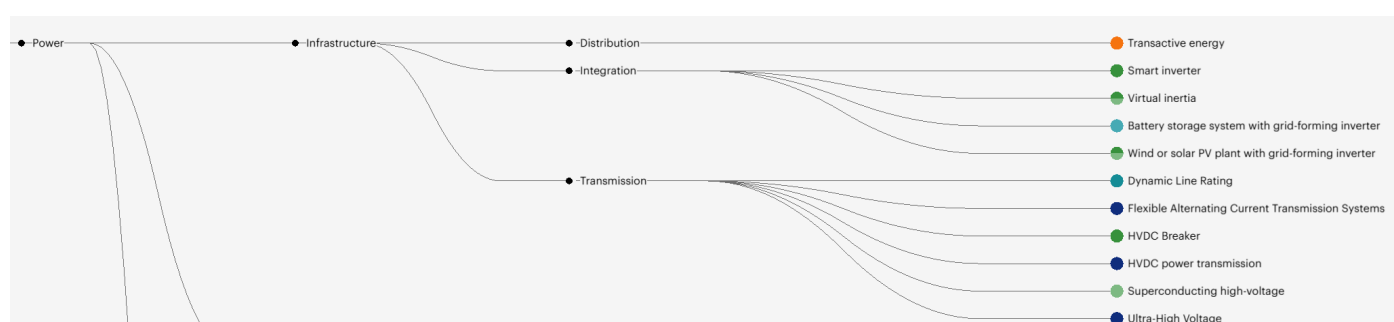


Source: [ETP Clean Energy Technology Guide – Data Tools - IEA](#)

Storage technologies are also important, with batteries gaining popularity thanks to cost reduction, which are used in both on-grid and off-grid solar.

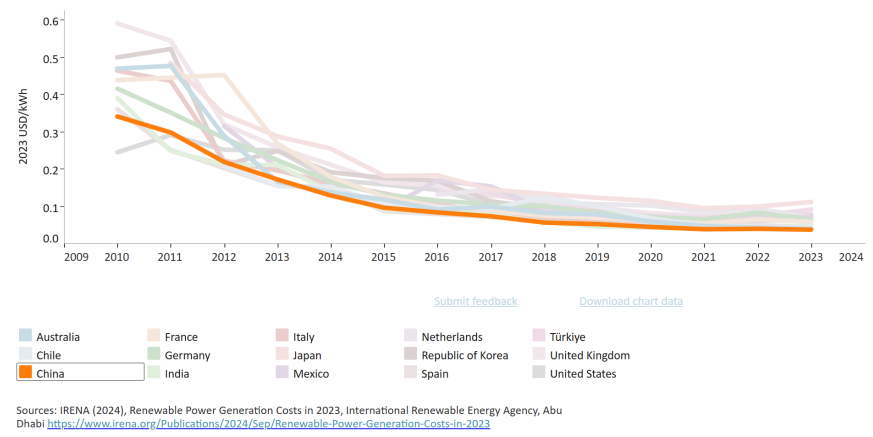
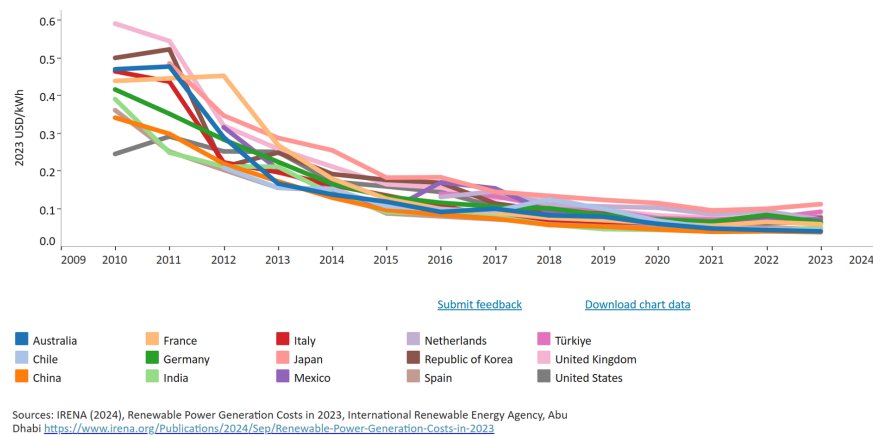


Source: [ETP Clean Energy Technology Guide – Data Tools - IEA](#)

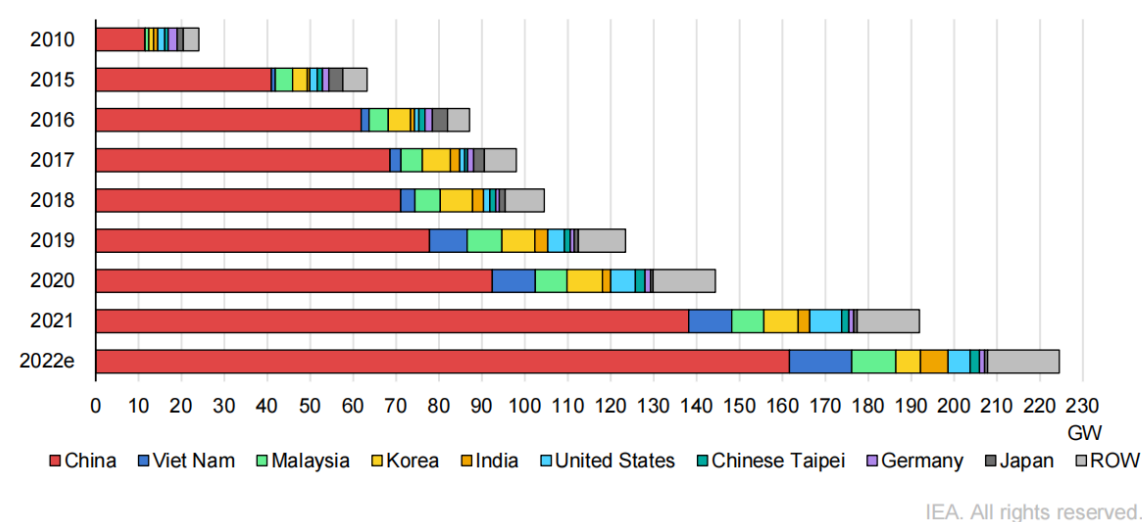


3 China's cost reduction in Solar PV is significant, making it affordable for developing countries

3.1 Cost reduction and China's dominance in all steps of production



Global solar PV module production, 2010-2022



Notes: ROW = rest of world. Values for 2022 are estimates.

Source: IEA analysis based on BNEF (2022a), IEA PVPS, SPV Market Research, RTS Corporation and PV InfoLink.

3.2 Appropriate products for developing countries for its convenience to use, lower cost, and availability of solar resources:

E.x.1 Africa

Initially, solar power generation in Africa was envisioned primarily through Concentrated Solar Power (CSP) technology.

- **How does it work?** CSP works by utilizing large arrays of mirrors or lenses to concentrate sunlight onto a central receiver or heat-absorbing tubes. This concentrated heat generates high-temperature steam, which subsequently drives turbines to produce electricity. Ambitious projects such as the "Desertec Initiative" aimed at harnessing the immense solar potential of the Sahara Desert, proposing massive CSP installations designed not only to meet local energy demands but also to export renewable power to Europe.
- **What's the challenge?** However, CSP faced significant practical and logistical challenges. Firstly, CSP plants are typically located in arid desert regions far from major population centers, **requiring extensive and costly cross-border transmission infrastructure**. The Desertec plan, for example, involved complex transnational coordination and enormous investments in grid interconnections across multiple countries, which ultimately proved politically and economically infeasible. Additionally, CSP technology **demands substantial quantities of water** for cooling and mirror cleaning—resources that are particularly scarce in desert environments.

In contrast, photovoltaic (PV) technology offers advantages that have made solar energy more accessible across Africa. PV systems directly convert sunlight into electricity using solar cells without the need for moving parts or high levels of water consumption. This makes PV **simpler, more scalable, and easier to deploy** in diverse locations, ranging from large-scale utility solar farms near cities to small distributed off-grid systems in rural and remote regions. Furthermore, the **dramatic fall in PV panel prices**—by over 80% in the last decade—has drastically reduced installation costs, making solar energy economically viable for millions of households and businesses.

E.x.2 SIDS (small island developing countries) saw a shift of priorities towards more scalable and low-cost solar PV, especially after the repercussions of COVID-19 when financial constraint worsened and international engineers cannot

travel to operate

Solar photovoltaics (PV): Additions grew steadily – from a moderate figure of 570 megawatts (MW) in 2015 to a significant rise to 860 MW in 2023 due to **declining solar installation costs**, increased international funding and the **suitability of solar energy to decentralised energy needs** .

Wind Energy: Fluctuations were observed in wind energy...reflecting logistical challenges (e.g. **the difficulty of maintaining wind turbines on isolated islands**, a **reliance on consistent wind resources**. By 2023, wind growth stagnated, with capacity staying at the level observed in 2022

Bioenergy capacity In 2023, bioenergy suffered a significant decline of -260 MW (IEA, 2024), driven by **high feedstock costs** and **operational inefficiencies** that forced the decommissioning of projects .

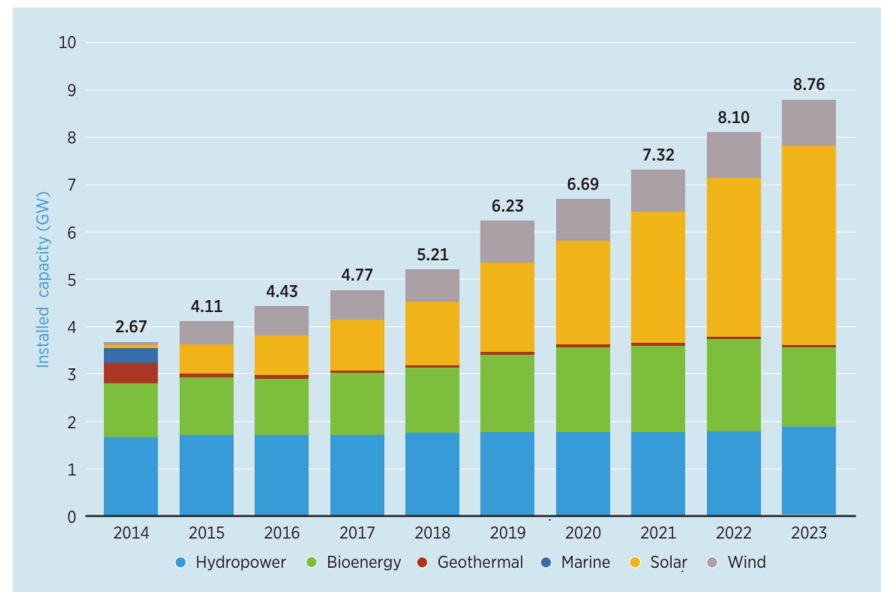
Hydropower was the most stable energy source, though with **natural limitations due to the small island setting** . Small-scale increases have occurred sporadically: 70 MW were added in 2023, after a few years of growth almost stagnating .

Geothermal and marine energy development declined sharply in 2015 amid **financial constraints...High upfront costs of feasibility and exploration, complex technical requirements and logistical challenges** deter most investments .

(Accumulative)

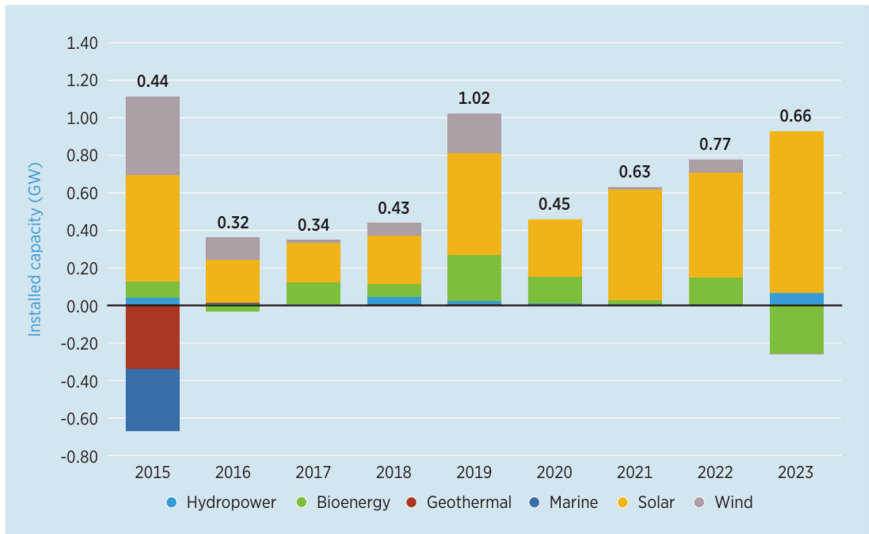
(Additional)

Figure 2 Total installed renewables-based electricity capacity, by technology, across all SIDS, 2014-2023



Source: IRENA, 2024.

Figure 3 Installed renewables-based electricity capacity across all SIDS, 2015-2023



Source: IRENA, 2024.

3.3 How does China reduce cost?

Key reasons driving down the cost are China's manufacturing base as "World Factory", big domestic market and effective industry policies.

- **Origins and Early Applications (1950s–1970s):** Solar photovoltaic technology was first developed in the mid-20th century, with innovation concentrated in EU, US and Japan. During this period, PV technology was primarily used in space applications due to its high cost and limited efficiency.
- **Initial Development and Export-Oriented Growth (1980s–2008):** Rapid policy-driven expansion in demand of solar PV emerged in Europe and North America, after the signing of international climate agreements (Kyoto Protocol). During this period, China became a key production center as world's manufacturing hub. However, China faced a **"dual external dependency"**—relying on imported raw materials (polysilicon, which is the most complex and capital-intensive part in pv cell, mainly produced in US and EU), and exporting most of its products, with minimal domestic deployment.
- **Domestic Policy Response and Market Development (2009–2012):** The 2008 financial crisis led to a decline in international demand, and Anti-dumping and countervailing duty investigations by the U.S. and EU further strained Chinese PV exports. In response, China implemented policies to stimulate domestic demand to support the PV industry and reduce reliance on exports.
- **Rapid Expansion and Technological Advancements(2013-):** China's solar PV industry has achieved remarkable growth through strategic industrial policies that promote large-scale production, foster industrial clusters, and encourage innovation. Government initiatives, including financial subsidies, feed-in tariffs, and programs such as "Golden Sun"(promoting solar in rural area), stimulated domestic demand and supported industrial expansion. The formation of integrated industrial clusters in provinces like Jiangsu and Anhui enhanced supply chain efficiency and

economies of scale. Consistent investment in R&D led to technological advancements in solar cell efficiency and manufacturing.

Three key factors behind China's cost reduction in solar PV are particularly noteworthy, reflecting structural characteristics that differentiate China from developed countries. First is **local-adapted manufacturing**, which prioritizes lower-cost technologies and substitutes imported machinery and production processes with domestic alternatives. Second is **learning by doing**, for which large-scale production is especially important. Third is the development of **industrial clusters**, a central component of China's industrial policy, which has also been increasingly mirrored in several ASEAN countries (Association of Southeast Asian Nations, members include Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam) in recent years.

- **Local-adapting manufacturing**

- (i) China's manufacturing target at low-cost technology, though may at the cost of efficiency loss**

Chinese scientists are working on essentially the same suite of emerging solar technologies as are scientists elsewhere in the world. However, when it comes to deciding which new solar technologies to manufacture,

China-based companies typically are more conservative than Western firms. Their main hesitation is cost.

Ex.1: One example is the decision by many of top China-based solar-module manufacturers to choose PERC rather than HIT or IBC (3 types of cell, with PERC less efficient than the other two) as the technology for their next-generation high-efficiency products. That is because, reconfiguring a traditional silicon-cell assembly line to produce PERC cells requires only two basic changes, far fewer than reconfiguring that line to produce HIT or IBC cells. So PERC cells cost less than HIT or IBC cells to produce.

Ex.2: Another example is that, Advanced Solar Power, a Chinese firm, chose to conduct only a few production steps in "clean rooms" (expensive facilities that minimize contamination to boost cell efficiency), whereas the U.S.-based First Solar, conducts its entire production process in clean rooms. This strategy is called "selective sophistication", by Dr. Wu, the starter of Advanced Solar Power. He estimated that First Solar modules on average are about 2 percentage points more efficient than Advanced Solar Power's but cost about 25% more.

Source: [2017-03-20-Stanford-China-Report.pdf](#)

- (ii) Domestic substitution of machines and production process**

Chinese manufacturing are using domestic substitutions for western devices in less sophisticated sections, for lower cost, easier usage, and quicker access to maintenance

Ex.1: Dr. Wu, Advanced Solar Power's chairman, said reconfiguring the tooling machines would have taken longer, cost more, and required more-difficult long-distance communications with the tooling manufacturer had the tooling come from a foreign supplier instead of from a China-based firm.

Ex.2: One cell manufacturer in China has four firing furnaces to make cells, with three of them made by Chinese firms, and one by a German company. The manager of the cell-production factory said the factory turns on the German furnace only if it has an order volume so large that it has maxed out the capacity of all three Chinese-made furnaces. The reason is that the factory's workers find it far easier to tweak the Chinese-made furnaces to increase the flexibility of production. By contrast, the German furnace, from a top supplier and of top quality, does not adjust to new production conditions easily. For example, the German-made one has its labels in English, which many assembly-line workers do not read. And while the Chinese-made machine uses a green light to indicate that the machine is in operation—green being intuitive in China, as in much of the world, to mean "go"—the German-made machine uses a red light for the same purpose.

Ex.3 In contrast, local device makers have the advantage to provide prompt feedback and repair service, thanks to industry cluster. Trina is one example. Ten of its suppliers' factories sit in the same industrial park as does Trina itself. The proximity allows engineers from Trina and its suppliers to meet to discuss their ongoing joint R&D projects on a weekly basis. When a piece of equipment breaks down, the equipment supplier commonly can send its maintenance team to diagnose the problem on the same day. The swift repair of equipment reduces machine downtime, which in turn increases the Trina factory's productivity.

Similar adaptation is also observed in EV manufacturing, with domestic machines, and production process redesigned to be more labor intensive.

In a report of BYD's initial development, it mentions that rather than using automated production lines, BYD replaced specialized machines with standard machines, automated machines with semi automated machines, and, wherever possible, imported machines with machines that were built in-house. One of the potential risks of this approach is that

BYD would not be able to maintain a consistent level of quality. To help resolve this problem, the company developed special tools and provided its workers with a program of education and training. Although the move away from an automated, capital-intensive production process could be seen as a backward step, BYD managers see this as part of an on-going learning process that takes into account local conditions and the low labor costs in China.

Source: [2017-03-20-Stanford-China-Report.pdf](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2322430) https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2322430

- **Learning by doing(LBD) from large-scale production**

LBD largely contributes to cost reduction, which can only be achieved with experience gained from mass production.

LBD is defined as the reduction in unit production cost as a function of cumulative experience in manufacturing industries. It emphasizes the subtle experience gained through operating and manufacturing process, different from technological progress and economics of scale.

In EV industry, the battery production process has at least 3 features that could contribute to LBD. First, the production process is highly complex and governed by hundreds of tuning parameters. The interconnected system needs to be **constantly fine-tuned and optimized using experience** to achieve efficiency, and this requires experience for there are subtle variations in different cells. Second, the production process is very sensitive to **material purity and requires stringent clean-room standards**, which also requires experience to operate even equipped with advanced facilities. Tiny amounts of impurities can cause high scrap and low yield rates. Third, the industry has been undergoing **continuous technological advances in new chemistry composition and production techniques**, which require learning and have important implications for production costs.

The contribution of LBD in EV battery is significant, accounting for a sizable 35.5% of the overall decline in battery costs from 2013 to 2020. Industry-wide technological progress was responsible for an additional 39.9% of the cost reduction, with the remainder explained by LBD in EV assembly, changes in battery chemistry, input cost fluctuations, and economies of scale. (Barwick et al., 2025 https://panlebarwick.github.io/papers/Yr25_LBD_BKLZ.pdf)

In Solar PV, LBD also matter, and there is an interesting example that companies put emphasize on encouraging workers to innovate in manufacturing process. A Chinese company, Yiling, launched a campaign called "grassroots-innovation program" to encourage assembly-line workers to harness their daily experience to suggest and implement ways to improve the line's efficiency and product quality. In one example, the operator of one of a wire-saw machine on a Yingli line suggested an adjustment to the machine that allowed the machine to slice silicon wafers more thinly and more uniformly.

- **Industry Cluster**

China's solar PV industry has achieved remarkable cost reductions and scale through strategic geographic and industrial clustering, particularly in the Yangtze River Delta region, which includes 3 provinces in China(Jiangsu, Zhejiang, and Shanghai). By 2016, this region alone accounted for 74% of China's solar cell capacity and 85% of its module capacity—equivalent to 45% and 60% of global production, respectively.

This regional concentration was enabled by large, state-supported industrial parks, typically established and managed by provincial or local governments. These parks cover vast areas—often thousands of acres—and offer integrated infrastructure, including highways, railways, power plants, apartment complexes, shopping centers, and manufacturing facilities. Crucially, they also provide companies with embedded government support staff who act as on-site industry consultants. These officials help firms navigate administrative procedures, cut through bureaucratic red tape, and accelerate plant setup—all while aiming to boost local employment and tax revenues.

By concentrating firms and services within these industrial parks, China has significantly reduced logistical and coordination costs: Proximity among firms has facilitated the rapid formation of a comprehensive domestic supply chain, reducing reliance on imported materials such as glass, EVA, and silver paste; In addition, the spatial clustering of equipment suppliers allows for timely maintenance and smoother coordination, further enhancing production efficiency and cost control.

4 What's the implications for other developing countries?

- **Appropriate product?**

Solar PV is very user-friendly, flexible in deployment and has no persistent requirement of resources input. However, it seems to be perfect product for both developed countries and developing countries, thus has no big mismatch on demand side. But it may suggest something why China invest in and adopt this type of product so much, not others

- **Appropriate Manufacturing? Appropriate FDI?**

China's local-adapted manufacturing and industry policy may offer a replicable model for other developing countries. Anecdotal evidence from solar industry experts suggests that China's industrial cluster pattern is being emulated in parts of Southeast Asia. And China's FDI in energy sector ramped up recently, especially in these countries, say solar photovoltaics in Vietnam and Malaysia, and new energy vehicles in Thailand and Brazil.

However, the impact of this FDI on the host countries' energy transitions seem to have mixed implications. On the one hand, Chinese investment brings much-needed capital, affordable equipment, local supply chain development and rapid deployment capacity. On the other hand, concerns remain regarding higher carbon intensity in China's manufacturing line, and lacks of meaningful local participation to build long-term capacity.

- **Appropriate Entrepreneurship?**

China's start-ups have attracted more than one-third of global early-stage energy venture capital. There might be a similar story in energy sector as your fantastic "appropriate entrepreneurship" paper.

- **Belt & Road Initiative cooperative countries**

China's green energy investments in BRI partner countries have grown rapidly in recent years, typically following four main cooperation models: EPC contracts, BOT arrangements, joint ventures, and localized manufacturing paired with project development.

- **EPC(engineering, procurement, and contract)**

Under EPC model, Chinese companies are contracted to design, procure, and construct renewable energy projects. This model is often used in host countries lacking technical capacity, as it enables rapid infrastructure deployment with minimal local participation in operations or ownership.

Key Examples:

- **Benban Solar Park, Egypt**China General Nuclear (CGN) was responsible for multiple EPC-contracted sub-projects within Egypt's 1.8 GW Benban Solar Park. Chinese firms delivered engineering services, modules, and labor, helping accelerate Egypt's solar build-out.
- **Garissa Solar Plant, Kenya**China Jiangxi Corporation for International Economic and Technical Cooperation (CJIC) built Kenya's largest solar plant (50 MW) under EPC. The project significantly boosted regional energy supply and was financed through a Chinese government concessional loan.
- **BOT (Build-Operate-Transfer)**

In BOT projects, Chinese firms finance, construct, and operate the project for a set period, then transfer ownership to the host country. This model suits large infrastructure with long payback periods and is common in hydropower and large-scale solar projects.

Key Examples:

- **Patuca III Hydropower Plant, Honduras**Although outside ASEAN, this project shows China's use of BOT globally. Built by Sinohydro, the plant was financed with Chinese loans and is operated by the Chinese contractor until transfer.
- **Solar PV Plant, Pakistan (Zonergy)**Part of the Quaid-e-Azam Solar Park, this project involves BOT elements with Chinese firm Zonergy building and operating a 300 MW segment under long-term agreements.
- **Lao National Grid Expansion**

EPC contracts by PowerChina helped build grid infrastructure to connect remote hydropower facilities to Laos' national grid, exporting excess electricity to Thailand and Vietnam.

- **Joint Venture (JV)** Joint ventures involve Chinese firms partnering with local companies or governments to share ownership, risks, and profits. This model helps ensure local policy alignment and may involve some local employment and technology transfer.

Key Examples:

- **LONGi Green Energy in Malaysia**LONGi formed a joint venture with a local partner in Kuching, Sarawak, to manufacture solar wafers and modules, integrating with local industrial parks and supply chains.
- **Zonergy and local utilities in Uganda**In solar mini-grid projects, Chinese firms have partnered with Ugandan energy authorities to co-develop and co-own solar installations, improving energy access.
- **Shanghai Electric + DEWA (Dubai)**For the 950 MW CSP-PV hybrid project in Dubai, Shanghai Electric formed a JV with the Dubai Electricity and Water Authority (DEWA), aligning Chinese technology with UAE energy strategy.
- **Localized Manufacturing + Project Development**

To bypass trade barriers and support industrial localization, Chinese companies have begun investing in local manufacturing in host countries, sometimes pairing it with downstream project development.

Key Examples:

- **BYD in Thailand (EV and Battery Plant)**In 2023, BYD began constructing an electric vehicle plant in Thailand's Rayong province to serve regional ASEAN markets. This includes battery assembly lines and potentially solar integration.
- **JinkoSolar in Malaysia and Vietnam**Jinko set up cell and module factories in both countries to serve overseas and regional markets, also developing solar farms in cooperation with local partners.
- **Oman Solar Component Manufacturing**Chinese investors have established a solar glass and module factory in Oman's Duqm SEZ, designed to supply large-scale PV projects in Gulf countries.

Relevant Dataset: [China Global Investment Tracker](#) | [American Enterprise Institute - AEI](#)

Appendix: a mind map of different types of solar

solar

